Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software engineer pursuing an MSCS at USC with 2+ years building **API-first platforms**, **React interfaces**, and AI-enabled products, with strong experience translating technical concepts into clear guidance, documentation, prototypes, and cross-functional delivery plans.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Designed a multi-tenant content and data platform for 500+ users, reducing latency from **1.2s to 350ms** by refining information structure, SQL access patterns, and backend workflows for better usability.
- Built automated validation and ETL checks for 5,000+ monthly jobs, improving implementation reliability and giving stakeholders faster feedback on regressions, data-quality drift, and operational issues.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed full-stack React and Flask workflows handling 10,000+ daily requests, translating instructor feedback into maintainable APIs, improved interfaces, and iterative enhancements that could be quickly validated in production.
- Built AI-assisted grading and rubric workflows with multimodal models and OCR, reducing feedback turnaround by **60%** while supporting clearer guidance and better operational handoffs for users.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built Python experimentation pipelines across 100,000+ trajectory records, enabling reproducible evaluation and clear technical artifacts that helped non-specialists understand tradeoffs, results, and implementation priorities.
- Produced technical reports and targeted test scenarios translating findings into actionable guidance, improving stakeholder alignment, reducing handoff ambiguity, and supporting better technical decision-making across teams.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered API and ETL microservices using Python, AWS Glue, and S3, improving response time by **30%** while supporting distributed integrations and structured delivery across customer-facing production workflows.
- Maintained 20+ automated test suites using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%** and strengthening release confidence across collaborative engineering and implementation teams.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed MERN and WebSocket modules for 2M+ synchronized records across eight workflows, improving responsiveness and helping cross-functional teams support real-time user needs with fewer stale-session issues.
- Automated Docker-based build and test flows with Jenkins, cutting release time by **50%**, improving deployment consistency, and enabling smoother validation and troubleshooting during project delivery.

Skills

Consulting & Delivery: technical discovery, requirements translation, stakeholder communication, documentation, workshops/support content, cross-functional collaboration
Web & APIs: JavaScript/TypeScript, React, Next.js, Node.js, Flask, FastAPI, REST APIs, API-first architectures
Data & Platforms: content/data modeling, PostgreSQL, MySQL, MongoDB, SQL, ETL pipelines, cloud workflows
Tools: Git/GitHub, Docker, Jenkins, AWS, GraphQL familiarity, Agile/Scrum, technical enablement

Projects

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, GCP*

- Built a secure document platform with semantic search, summarization, and collaborative Q&A, demonstrating structured content reuse and RAG-style workflows over 50K+ documents while reducing analysis time by **60%**.

PiXMod Live | *Next.js, MongoDB, Stripe, Transformers*

- Built an AI-powered Next.js SaaS platform for image editing and search, integrating reusable data models, Stripe billing, and MongoDB-backed workflows while improving premium conversion by **18%**.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Built an AI-powered SQL workspace supporting 10+ databases, creating a prototype-rich Java and TypeScript environment for schema exploration, query generation, and faster onboarding into complex data workflows.